

Introduction to GitHub

GitHub is a platform for hosting and collaborating on code projects. It uses Git, a version control system, to track

1. Key Concepts

- Repository (Repo): A storage location for your project files and history.
- Commit: A snapshot of your project at a specific point in time.
- Branch: A parallel version of your code for adding features or making changes.
- Pull Request (PR): A request to merge changes from one branch into another.
- Clone: A local copy of a repository.
- Push: Upload local changes to GitHub.
- Pull: Download changes from GitHub to your local copy.

2. Setting Up GitHub

1. Sign up at <https://github.com> and create an account.
2. Install Git on your computer (<https://git-scm.com/downloads>).
3. Configure Git with your username and email:
`git config --global user.name "Your Name"`
`git config --global user.email "you@example.com"`

3. Creating a Repository

1. Click the '+' icon on GitHub and select 'New repository'.
2. Name your repository and choose visibility (Public/Private).
3. Optionally initialize with a README file.
4. Click 'Create repository'.

4. Cloning a Repository

1. Navigate to your repository on GitHub.
2. Click the green 'Code' button and copy the HTTPS URL.
3. In your terminal, run:
`git clone <URL>`
4. This creates a local copy of your repository.

5. Making Changes and Committing

1. Open files in your local repository and make changes.
2. Stage changes:
`git add <file>`
or add all changes:
`git add .`
3. Commit changes:
`git commit -m "Description of changes"`

6. Pushing Changes to GitHub

1. Push your committed changes to GitHub:
`git push origin main`
(Replace 'main' with your branch name if different.)

7. Working with Branches

1. Create a branch:
`git checkout -b new-feature`
2. Switch branches:
`git checkout main`
3. Merge changes:
`git merge new-feature`

8. Pull Requests

1. Push your branch to GitHub.
2. On GitHub, open a Pull Request to merge it into 'main'.
3. Team members can review and approve before merging.

9. Syncing Changes

1. Pull the latest changes from GitHub:
`git pull origin main`
2. Resolve any merge conflicts by editing files, staging, and committing.

10. Tips for Using GitHub Effectively

- Commit often with clear messages.
- Use branches for features and bug fixes.
- Pull changes frequently to stay in sync.
- Use Pull Requests for collaborative review.
- Keep your main branch stable.